ADAR'S STUDY SERIES

LINEAR ALGEBRA

PART 2 · ECHELON FORMS & GENERAL SOLUTIONS

Echelon Form · RREF · Pivots & Free Variables · General Solutions · Consistency Cases

Deep Violet & Amber Edition

CONTENTS

1 · ECHELON FORM (REF)

A matrix is in echelon form if it satisfies the following two conditions:

- All non-zero rows are above any zero rows.
- Each non-zero leading entry (the first non-zero number from the left in a row) is to the left of the non-zero leading entries of the rows below it.

Notation Key

- ■ or 1 = any non-zero number
- * = any number
- 0 = zero entry

Examples of Echelon Form

2 x 3 Example

```
( ■  *  * )
( 0  0  ■ )
```

4 x 6 Example

```
( 0  ■  *  *  *  * )
( 0  0  0  ■  *  * )
( 0  0  0  0  0  ■ )
( 0  0  0  0  0  0 )
```

2 · REDUCED ROW ECHELON FORM (RREF)

A matrix is in reduced row echelon form if it meets the criteria for echelon form, and additionally satisfies:

- Every non-zero leading entry (pivot) equals 1.
- There are zeros above each non-zero leading entry (pivot), as well as below it.

Examples of RREF

2 x 3 Example

```
( 1  *  0 )
( 0  0  1 )
```

4 x 6 Example

```
( 0  1  *  0  *  0 )
( 0  0  0  1  *  0 )
( 0  0  0  0  0  1 )
( 0  0  0  0  0  0 )
```

3 · KEY THEOREMS

- **Uniqueness: Any matrix can be transformed into a unique reduced row echelon form using elementary row operations.**
- **Invariance: Elementary row operations never change the solution set of the underlying linear system.**

4 · PIVOT CONCEPTS & TERMINOLOGY

Once a matrix is in echelon form, the following terms describe its structure:

- **Pivot → The non-zero leading entry of a row.**
- **Pivot Column → A column that contains a pivot (non-zero leading entry).**
- **Pivot Variable → The variable corresponding to a pivot column.**
- **Free Column → A non-pivot column in the coefficient matrix (excluding the augmented column).**
- **Free Variable → The variable corresponding to a non-pivot column.**

> *Note for augmented columns: The final column of an augmented matrix (the right hand constants) is NOT a free column — it does not belong to the coefficient matrix or correspond to any variable.*

5 · GENERAL METHODS FOR ROW REDUCTION

Step 1 — Forward Pass (Top-Left to Bottom-Right)

Work from top-left to bottom-right to bring the matrix into Echelon Form (REF). Use row operations to reduce the matrix into Echelon Form (REF)

Step 2 — Backward Pass (Bottom-Right to Top-Left)

Work from bottom-right to top-left to bring the matrix into Reduced Row Echelon Form (RREF). Work backward to create zeros above each pivot and scale pivots to 1 to achieve Reduced Row Echelon Form (RREF).

6 · WORKED EXAMPLE — FULL REDUCTION

Put the following matrix into reduced row echelon form, then find the general solution.

Starting Matrix

$$\begin{pmatrix} 0 & 0 & 1 & 4 & -1 & 1 \\ 2 & -2 & 4 & 6 & 3 & 7 \\ 1 & -1 & 1 & -1 & 2 & 1 \end{pmatrix}$$

Solution:

Part (a) — Row Operations to RREF

Step 1 — Forward Pass (to Echelon Form)

Swap R1 and R3 (R1 ↔ R3):

$$\begin{pmatrix} 1 & -1 & 1 & -1 & 2 & 1 \\ 2 & -2 & 4 & 6 & 3 & 7 \\ 0 & 0 & 1 & 4 & -1 & 1 \end{pmatrix}$$

Eliminate R2's leading entry (-2R1 + R2 → R2):

$$\begin{pmatrix} 1 & -1 & 1 & -1 & 2 & 1 \\ 0 & 0 & 2 & 8 & -1 & 5 \\ 0 & 0 & 1 & 4 & -1 & 1 \end{pmatrix}$$

Swap R2 and R3 (R2 ↔ R3):

$$\begin{pmatrix} 1 & -1 & 1 & -1 & 2 & 1 \\ 0 & 0 & 1 & 4 & -1 & 1 \\ 0 & 0 & 2 & 8 & -1 & 5 \end{pmatrix}$$

Eliminate entry below the pivot (R3 - 2R2 → R3):

Echelon Form (REF)

$$\begin{pmatrix} 1 & -1 & 1 & -1 & 2 & 1 \\ 0 & 0 & 1 & 4 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{pmatrix}$$

Step 2 — Backward Pass (to RREF)

Clear entries above the pivot in Column 5 (R2 + R3 → R2) and (R1 - 2R3 → R1):

$$\begin{pmatrix} 1 & -1 & 1 & -1 & 0 & -5 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{pmatrix}$$

Clear the entry above the pivot in Column 3 (R1 - R2 → R1):

Reduced Row Echelon Form (RREF)

$$\begin{pmatrix} 1 & -1 & 0 & -5 & 0 & -9 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{pmatrix}$$

7 · WORKED EXAMPLE — GENERAL SOLUTION

From the RREF augmented matrix:

$$\left(\begin{array}{ccccc|c} 1 & -1 & 0 & -5 & 0 & -9 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right)$$

Identifying the Variables

- **Pivot Columns: 1, 3, 5 → Pivot Variables: x_1, x_3, x_5**
- **Free Columns: 2, 4 → Free Variables: x_2, x_4**

System Equations

$$x_1 - x_2 - 5x_4 = -9$$
$$x_3 + 4x_4 = 4$$
$$x_5 = 3$$

General Solution

$$x_1 = x_2 + 5x_4 - 9$$
$$x_2 = \text{free}$$
$$x_3 = -4x_4 + 4$$
$$x_4 = \text{free}$$
$$x_5 = 3$$

8 · CONSISTENT vs. INCONSISTENT SYSTEMS

- **Consistent System → a linear system that has at least one solution (unique or infinitely many).**
- **Inconsistent System → a linear system that has no solutions.**

When reducing an augmented matrix [A | b], exactly one of three cases will apply — covered in the next sections.

9 · CASE 1 — NO SOLUTION (INCONSISTENT)

Condition

- **The last (constant) column contains a pivot.**
- **Equivalently: a row of the form [0 0 … 0 | ■] exists, with ■ ≠ 0.**

Reason: such a row translates to the equation 0 = ■, which is logically impossible.

Example

Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & 0 & 7 & -12 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 5 \end{array}\right]$$

$$x_1 + 7x_3 = -12$$

$$x_3 = 7$$

$$0 = 5 \quad \text{<-- Impossible!}$$

Conclusion: No Solution (Inconsistent)

10 · CASE 2 — UNIQUE SOLUTION (CONSISTENT)

Condition

- **The last column is NOT a pivot column, AND**
- **There are NO free variables — every coefficient column has a pivot.**

Example

$$\left[\begin{array}{cc|c} 1 & -1 & 4 \\ 0 & 5 & 15 \end{array}\right]$$

Method 1 — Back-Substitution from Echelon Form

$$\text{Row 2:} \quad 5x_2 = 15 \quad \Rightarrow x_2 = 3$$

$$\text{Row 1:} \quad x_1 - x_2 = 4 \quad \Rightarrow x_1 - 3 = 4 \quad \Rightarrow x_1 = 7$$

Method 2 — Reduce Fully to RREF First

Divide Row 2 by 5 (R2 / 5 → R2):

$$\left[\begin{array}{cc|c} 1 & -1 & 4 \\ 0 & 1 & 3 \end{array}\right]$$

Add R2 to R1 (R1 + R2 → R1):

$$\left[\begin{array}{cc|c} 1 & 0 & 7 \\ 0 & 1 & 3 \end{array}\right]$$

Unique Solution: $x_1 = 7, \quad x_2 = 3$

11 · CASE 3 — INFINITELY MANY SOLUTIONS

Condition

- **The last column is NOT a pivot column, AND**
- **At least one free variable exists — at least one coefficient column has no pivot.**

Example — RREF Matrix

$$\left[\begin{array}{ccccc|c} 1 & -1 & 0 & -5 & 0 & -9 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right]$$

1. Identify Variables

- Pivot Columns: 1, 3, 5 → Pivot Variables: x_1, x_3, x_5
- Free Columns: 2, 4 → Free Variables: x_2, x_4

2. System Equations

$$x_1 - x_2 - 5x_4 = -9$$

$$x_3 + 4x_4 = 4$$

$$x_5 = 3$$

3. General Solution (Isolate Pivot Variables)

$$x_1 = x_2 + 5x_4 - 9$$
$$x_2 = \text{free}$$
$$x_3 = -4x_4 + 4$$
$$x_4 = \text{free}$$
$$x_5 = 3$$

Particular Solution

Setting the free variables $x_2 = 0$ and $x_4 = 0$ gives one specific solution vector:

$$(x_1, x_2, x_3, x_4, x_5) = (-9, 0, 4, 0, 3)$$

12 · QUICK DECISION SUMMARY TABLE

Last Column Has a Pivot?	Free Variables Present?	System Status	Number of Solutions
Yes [0 ... 0 | ■]	N/A	Inconsistent	0 Solutions
No	No	Consistent	1 Unique Solution
No	Yes	Consistent	Infinitely Many

13 · KEY POINTS & SUMMARY

- **Echelon Form (REF) requires non-zero rows on top and staircase-pattern leading entries.**
- **RREF additionally requires all pivots to equal 1, with zeros above AND below each pivot.**
- **Every matrix has a unique RREF, reachable through elementary row operations.**
- **Pivot columns fix pivot variables; non-pivot columns give free variables.**
- **Reduction proceeds in two passes: forward to REF, then backward to RREF.**
- **A pivot in the constants column always signals an inconsistent (no-solution) system.**
- **No free variables and a consistent system → exactly one unique solution.**
- **At least one free variable and a consistent system → infinitely many solutions.**

END OF PART 2 · NEXT: PART 3 — VECTORS & VECTOR SPACES